

Genetics of Trinucleotide Diseases

This Land Is  Your Land

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Learning Objectives

- Explain how trinucleotide repeat diseases mutate and why that gives rise to anticipation
- Explain the concepts of premutation and full mutation, explain how length of such repeat structures would normally be measured, and be able to interpret electrophoresis of such results
- State that CAG repeats lead to polyQ amino acid areas and explain the hypothesis of sequestration of proteins with affinity for such areas leading to adverse effects in the cell
- Express that other repeats are usually found outside the coding region and explain examples of how that leads to disease
- State and apply that most trinucleotide diseases are inherited in a dominant fashion, that Fragile X syndrome fits neither dominant nor recessive, and that Friedreich Ataxia is autosomal recessive
- Explain the two different mechanisms involved in (1) fragile X syndrome and (2) the diseases associated with premutation in the same gene
- Explain genetic counseling aspects of these diseases
- **Recommended reading: Emery's elements of medical genetics pp241-242, 273-275, 285-286**

A very fundamental observation

1. Effect of mutation on quantity of gene product – and consequence



Increased → usually bad

Unchanged → sometimes OK

Decreased → usually bad

2. Change quality of gene product → change quantity of other gene products → usually just as bad

These relationships are things you should know already. It is a framework that is important for understanding what is going on in trinucleotide (and other) diseases.

Bullet 1 is about changes to quantity of protein, bullet two is about change to quality to gene product. To remind you what you have learned before: Sickle cell disease is change to quality of gene product (one amino acid changed to another but no change in quantity) while Thalassemia is result of diminished amount of gene product.

Tri-nucleotide (triplet) repeats

- Short (few repeat units): stable
- Medium: unstable, but no(?) symptoms
- = **premutation**
- Long (many repeat units): unstable and disease causing
- = **full mutation**
- Instability → anticipation
 - in meiosis, often from one parent only

The definitions of premutation, full mutation and anticipation must be understood (not only memorized). Several examples will be discussed.

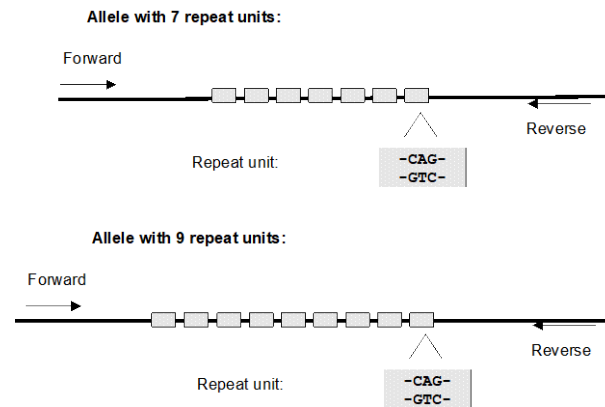
For premutation, there is at least one example where premutation carry risk of its own sets of symptoms, and where the mechanism is different for premutation and for full mutation

Slide 5



I think the rest of the lecture is easier to understand if this is understood. We hear that Step 1 often test on mechanisms, so this might also be important for that reason.

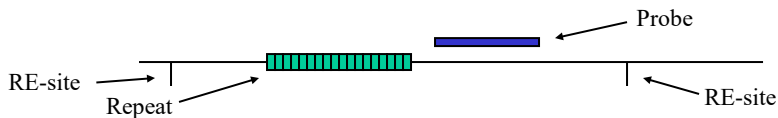
Determining the number of repeat units with PCR



Note for both this and the following slide: I expect that you by now are so familiar with electrophoresis that I do not have to show specific examples of each of these. It will be fair game for exams.

Determining the length of a repeat (2)

- Problem with PCR is that it is very difficult to amplify alleles with more than maybe 250 repeats
- Alternative is Southern blot using an enzyme with flanking cut-sites and a probe binding to a unique DNA sequence between the cut-site and the repeat



When the length of the repeat varies, so does the length of the overall fragment.

Repeat types and for type I: Huntington
disease

Type I vs Type II repeats

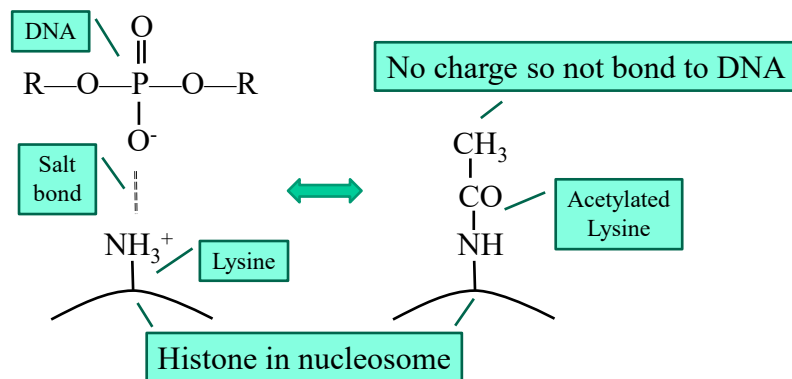
	Type I	Type II
Repeat unit	CAG	Several different
Amino acid encoded	Glutamine (Gln)	None (outside coding region)
Full mutation	Shorter	Can be very long
Affected body part	Brain	Several
Diseases (not complete)	Huntington, Spinocerebellar ataxia types 1 to N	Friedrich Ataxia, Fragile X syndrome, Dystrophia Myotonica

Poly-glutamines (polyQ) (from type I repeats)

- Interact with other proteins
 - Frequently found in regulators of transcription
- When expanded:
 - Interact with additional proteins
 - Forms aggregates → sequester other proteins (hindering normal function) → precipitation
 - Acetyl transferases are one such class of protein that is attracted to poly-Q

The bullet about sequestration and precipitation essentially is the same as hindering these other proteins in doing their job.

Histone Lysine and Acetylation – normal function



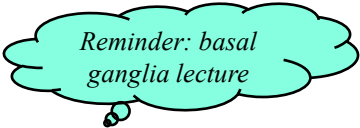
The drawing illustrates decreasing the binding affinity of DNA to the histone.

Another possible mechanism is proteins binding to acetylated lysine and changing how the region of DNA is regulated.

When the histone acetylase is sequestered by polyQ protein it cannot add the acetyl groups and the above mentioned changes do not happen.

Huntington's Disease symptoms

- Progressive
 - tremor
 - chorea
 - dementia
 - psychiatric symptoms
- Onset variable: 10% before age 20, peak onset age 35-50, a few very late
- Death (on average): 17 years after onset
- Variable expressivity: age of onset varies
- Inheritance: autosomal dominant (high penetrance)
- Anticipation occurs (see later)



*Reminder: basal
ganglia lecture*

Huntington's Disease

- Repeat type: CAG → poly-glutamine (type I)
- Repeat length
 - 10-29 repeats: normal (stable)
 - 30-35 repeats: premutation
 - 36-121 repeats: full mutation
 - (incomplete penetrance with 36-40 repeats)
- Instability in **male** gametogenesis
- Normal quantity/abnormal quality of Huntingtin

Huntingtin is the name of the protein encoded by the gene that can be come mutated and cause Huntington disease. The change in quality of the protein then leads to abnormal quantity in other proteins.

Notice that instability is only in male meiosis, repeats seems to be stable in female gametogenesis.

Anecdotally: a person is known to have had 60 repeats and not have developed any symptoms into her 60ties. She would, based on the repeat numbers, be expected to have developed the disease much earlier. Therefore, the disease may not have quite as high penetrance as previously described (99% instead of 100% for long repeats?).

Why do the long repeats cause symptoms?

- Researchers still disagree
- Some ideas backed by available evidence:
 - Changes to acetylation as previously seen
 - Sequestration of CREB and other transcription factors
 - Changes to expression of several genes such as BDNF
 - Interference with system for protein breakdown
 - Interference with vesicle transport to nerve synapses (including transport of BDNF)
 - BDNF has several important roles in the brain including roles in memory and neurogenesis

CREB = **cAMP response element-binding protein**, a transcription factor

BDNF = **Brain-derived neurotrophic factor**

The truth to why there are problems might be the sum of several different mechanisms, it does not have to be only one.

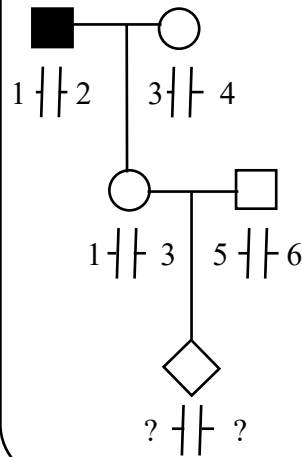
Problem

- Mrs. Jones wants a child.
- Mrs. Jones has one parent with Huntington.
- Mrs. Jones does not want to know her own carrier status, but she also does not want to have a child with the disease.
- *How can you help her?*

Solution

Marker analysis –
preimplantation diagnosis:

- If the embryo contains the allele from the grandfather (1): the embryo will be not be used; If the embryo contains the allele from the grandmother (3): embryo will be placed in uterus
- Elective abortion instead of preimplantation



Whether one, the other, or both methods (as described in the bullets) are acceptable to the prospective mother is her decision. As of recently, abortion might be illegal in some areas.

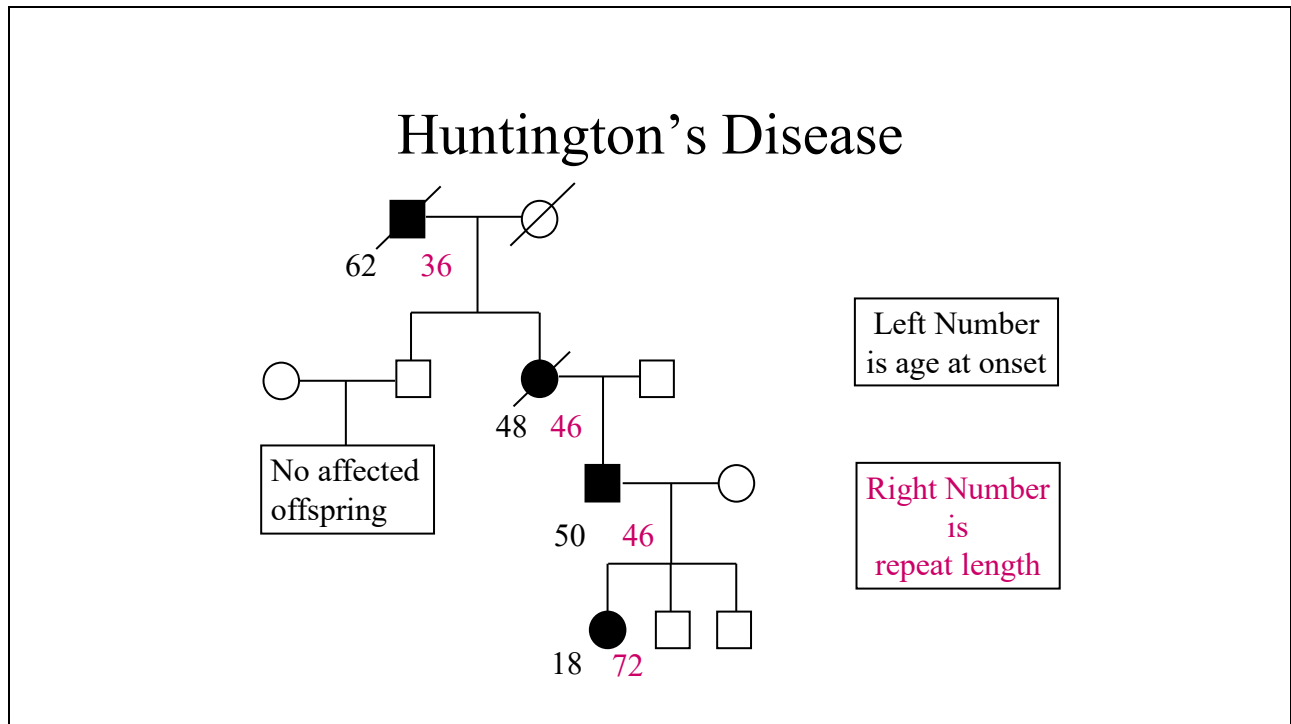
Huntington's Disease

- Odd fact: homozygotes for this dominant disease are no worse off than the heterozygotes
- One of the lowest mutation rates of any known locus (for repeats under 20 units)
- Well known Country & Western singer Woody Guthrie died from this disease



From <http://www.oktourism.com/>

Reminder: for most dominant diseases, homozygotes are significantly more affected than heterozygotes.



Repeat numbers increase when progressing down the pedigree. That is called **anticipation**. As a result of the longer repeats, the age of onset is earlier/symptom severity worse. Also notice that for Huntington disease, change in repeat number is expected in male meiosis but not in female meiosis.

Anticipation

- Describes earlier onset and/or more severe symptoms in later generations
- The anticipation in triplet repeat diseases is a result of two factors:
 - The repeat number often increases in meiosis, only or predominantly in one sex.
 - Increased repeat number correlates with more severe phenotype/earlier onset.

Diseases with type II repeats

Fragile X syndrome
Myotonic Dystrophy
Friedreich Ataxia

Fragile X also often is called FRAXA

There are other diseases of type II, but these are the most likely to occur again in your teaching and in the step exams.

FMR1 gene mutations

(FMRP TRANSLATIONAL REGULATOR 1)

- Repeat unit: CGG = $5'\text{-CGGCGGCGG-3'}$
 $3'\text{-GCCGCCGCC-5'}$
- Location: 5'UTR
- Involvement of methylation $5'\text{-CGGCGGCGG-3'}$
 $3'\text{-GCCGCCGCC-5'}$
(nucleotides in red might be methylated)
- Gene product FMRP
- In full mutation: normal quality, reduced quantity

Notice that the pattern of methylation is symmetrical, if a C is methylated on one strand then there is another C on the other strand that is also methylated. Please look at the Imprinting lecture for effects of methylation.

Repeat range	Designation	Symptoms	Penetrance males	Penetrance females
5-44	Normal	None		
45-54	Intermediate	None detected	-	Some expansion to premutation (only) in meiosis
55-200	Premutation	FXTAS, FXPOI	20-46% NA	Less frequent than men 15-20% - dep. on repeat length
>200 – methylated (if unmethylated more like premutation)	Full mutation	FXS	80-100% In meiosis, only transmission of premutation alleles	<50% for intellectual disability, higher for more subtle signs 50% transmission of FM in meiosis

FXS = Fragile X Syndrome

FXTAS = Fragile X-Associated Tremor/Ataxia Syndrome

FXPOI = Fragile X-Associated Primary Ovarian Insufficiency

Notice that an allele passed on from a male with FSX was a full mutation in him but can only be a shortened, premutation version in an offspring.

FXS = Fragile X syndrome

FMR1 alleles in the full mutation range

- Long face, large ears, prominent jaw, prominent forehead, strabismus (cleft palate), macrocephaly
- Hyperflexible joints, mitral valve prolaps
- Macroorchidism (>25 ml)
- Mild to severe intellectual disability, autism spectrum disorders
- Seizures
- Attention deficit, hyperactivity, anxiety
- Stereotypic movements
- Hand biting
- Gaze aversion
- **Variable**
- Males more strongly affected than females
- Prevalence 1/4000 males, 1/8000 females



As stated, the presentation is variable and it is unlikely that any single patient has everything mentioned here.

FXTAS = Fragile X-Associated Tremor/Ataxia Syndrome

- Onset > 50 years-of-age
- Progressive
- Intention tremor, cerebellar ataxia, peripheral neuropathy, atypical parkinsonism, and dementia
- white matter lesions involving middle cerebellar peduncles (MRI)
- Declines in intellect, short-term memory, and executive functioning
- Possible: personality or mood changes,
- increased irritability, and impulsive behavior
- **FMR1 alleles in the premutation range**
- **Incidence 1/1000 males (?)**

FXPOI = Fragile X-Associated Primary Ovarian Insufficiency

- Diminished ovarian reserve → irregular menses
- Elevated FSH
- Reduced fertility
- Menopause before age 40
- FXPOI cause 4-6% of all POI
- **FMR1 alleles in the premutation range**
- 15-20% of those women get FXPOI
- Risk highest around 100 repeats, less both above and below
- FXPOI not seen with full mutations.

About 1/200 women should have alleles in the range causing the risk. About 25% of those develop POI. Statistics come from <https://ghr.nlm.nih.gov/condition/fragile-x-associated-primary-ovarian-insufficiency#statistics>

FMR1 expression

Premutation repeats:

- mRNA increased
- Increased transcription
- RNA toxicity
 - Sequestration of mRNA binding proteins
→ inclusion bodies

Full mutation:

- mRNA decreased to 0
- Long repeat attracts methylation → attracts proteins → inactivates promoter → reduced transcription

This part similar to imprinting

Notice the opposite effect on mRNA levels and therefore the completely different mechanisms downstream from that.

FMR1 Protein (FMRP)

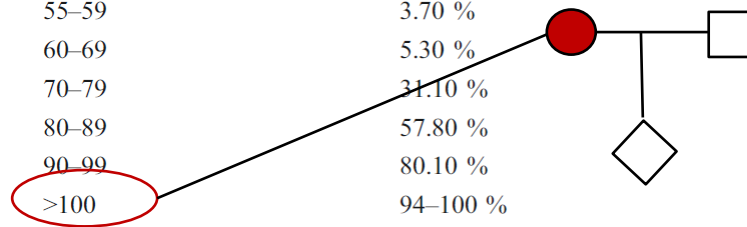
- Binds mRNA
- Inhibits translation
 - Probably regulated
- If absent → dysregulation of translation

Involved with metabotropic glutamate receptor (mGluR) signaling among others

Risk for repeat expansion depends on repeat size

Table 1 Transmission of the premutation in females

Maternal Repeat Size	Risk of Expansion to Full Mutation
55–59	3.70 %
60–69	5.30 %
70–79	31.10 %
80–89	57.80 %
90–99	80.10 %
>100	94–100 %



*Nolin et al. Am J Hum Genet 2003; 72: 454–464

From <https://fragilex.org/wp-content/uploads/2012/01/Genetic-Counseling-and-Testing-for-FMR1-Gene-Mutations-Practice-Guidelines-of-the-National-Society-of-Genetic-Counselors.pdf>

Notice that in the range of premutation size where FXPOI is most likely, repeat expansion is almost certain if that allele is passed on (remember that there are two X chromosomes so it could be the other allele that is passed on).

One important distinction is about interrupted repeats versus un-interrupted repeats. An interrupted repeat is more stable. The interruptions are of the form that every 10th repeat unit (approximately) is AGG instead of being CGG.

Myotonic Dystrophy = Dystrophia Myotonica



- Myotonia – distal to proximal, including facial, grip lock
- Muscular dystrophy
- Cataracts
- Cardiac conduction abnormalities
- Infertility
- Insulin resistance
- Congenital form: development abnormalities
- 1/8000

Grandmother and mother around 100 repeats, infant >1000

More on the clinical presentation: https://www-uptodate-com.rossuniversity.idm.oclc.org/contents/myotonic-dystrophy-etiology-clinical-features-and-diagnosis?search=myotonic%20dystrophy&source=search_result&selectedTitle=1~84&usage_type=default&display_rank=1

Types of Myotonic Dystrophy DM1 vs DM2

- Type 1 – CTG repeat in 3' UTR of DMPK
- Normal 5-34 / premutation 35-49 / full mutation >50 (to thousands)
- Type 2 – CCTG repeat in intron 1 of ZNF9
- Normal 11-26 / premutation n.d. / full mutation >75 (to thousands)

Notice similarity: CTG vs CCTG

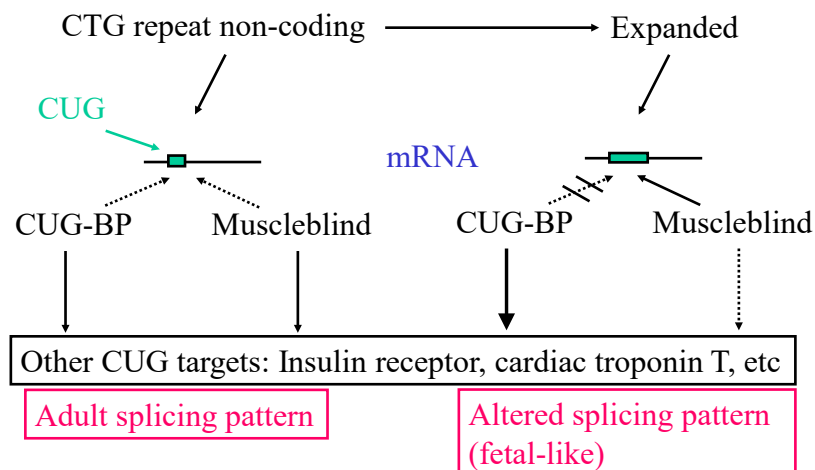
DMPK = Dystrophia Myotonica Protein Kinase

ZNF9 = zinc finger protein 9

DMPK = Dystrophia Myotonica Protein Kinase

ZNF9 = zinc finger protein 9

Pathogenesis of Dystrophia Myotonica (DM1 or DM2)



In the normal situation, normal amounts of both CUG-BP and Muscleblind proteins are available for regulating splicing in the cell. With abnormally long CTG repeats being transcribed, binding of Muscleblind protein to this mRNA is increased and of CUG-BP decreased, meaning an abnormal (fetal-like) balance in availability of the two proteins to regulate splicing of other targets.

More details about those targets in upcoming slide.

Caption of previous slide

- In normal adults, muscleblind influences splicing of at least 13 mRNAs
- In person with expanded CTG (or CCTG) in the DM gene, muscleblind gets sequestered by the DM mRNA
- That gives access for the CUG-BP protein to influence splicing into a fetal pattern for those 13 mRNAs

Pathogenesis of Dystrophia Myotonica (DM1 or DM2)

- 13 different genes expressed in embryonic form due to aberrant splicing
- Includes Insulin receptor, cardiac troponin T, etc.
 - Insulin receptor → embryonic form → weaker signal → insulin resistance
 - Muscle-specific chloride channel → loss of CIC1 mRNA → myotonia

Freidrich Ataxia

- Presents with ataxia
 - Usual onset 5-15 (def. <25) years of age
- Progressive:
 - **degenerative** atrophy of the posterior columns of the spinal cord
 - Loss of reflexes
 - Loss of motor strength
 - Usually results in need for wheelchair
 - Sensory loss
 - Dysarthria
 - Cardiomyopathy
 - This and other heart abnormalities result in premature death
 - Kyphoscoliosis, foot deformities
- A variety of complications can be seen
- 1 in 29,000 white people (data on other populations?)



From Uptodate: Ataxia can be defined as a disturbance in the smooth, accurate coordination of movements. It is most commonly manifested as an unsteady gait

Dysarthria: loss of pronunciation

Kyphoscoliosis: dual direction scoliosis

FA Inheritance

- Autosomal recessive!
- GAA repeat in first intron of frataxin (FXN) gene (9q13)
- Normal/premutation/full mutation: 7-34/35-66/66-1700 (most 600-900)
 - reduced penetrance below >100
 - Interrupted repeats more stable (e.g., GAT unit intermixed into the GAA repeat)
- Loss of function (hinders transcription)
- DNA testing is part of diagnostic criteria

Autosomal recessive, loss of function mutation because mRNA is not transcribed and therefore, protein is not produced.

About abbreviations: FRDA is today most often used as an abbreviation for the disease while FXN is used as an abbreviation for the name of the gene and the protein, Frataxin. However, it is also possible to find FRDA used as abbreviation for the gene.

How is transcription reduced in FA?

Likely two mechanisms:

- A long GAA repeat leads to epigenetic silencing of (or at least lowers output from) the promoter
- The presence of a long repeat in intron 1 interferes with splicing, further lowering mRNA production

Reference for these mechanisms: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2924884/>

Molecular effects of lacking frataxin protein (still hypothetical)

- Frataxin is localized to mitochondria
- Involved in mitochondrial iron metabolism as well as loading iron into FeS proteins
- Loss of this protein → mitochondrial iron buildup → oxidative damage
- But also malfunction of electron transport chain (FeS)

Frataxin is most highly expressed in tissues needing a high metabolic rate.